

END TERM EXAMINATION

THIRD SEMESTER [B.TECH] JANUARY 2024

Paper Code: AIML/IOT/AIDS-201

Subject: Data Structures

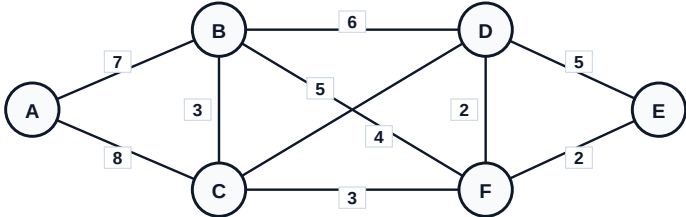
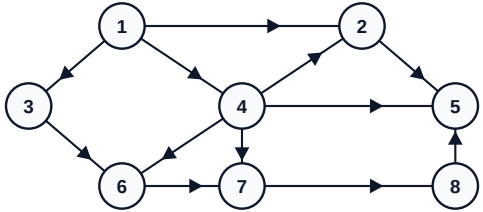
Time: 3 Hours

Maximum Marks: 75

Note: Attempt five questions in all including Q.No. 1 which is compulsory. Select one question from each unit. Assume missing data if any.

Q. No.	Question	Marks
Q1 (i)	What is Data Structures? Why do we need data structures? In how many ways can you classify the data structures?	3
(ii)	Explain features of list, set, tuples, and dictionary data structures.	3
(iii)	Given a sorted list of numbers {1, 13, 19, 25, 38, 55, 70, 80}. Search for value 70 using Binary Search Algorithm.	3
(iv)	Explain threaded binary tree?	3
(v)	Explain Topological sort with the help of example.	3
UNIT – I		
Q2 (a)	Write an algorithm/function to convert the infix expression into prefix expression using stack. Using the above algorithm convert the following infix expression into prefix using stack: P * O / R + (S - T * U)	7
(b)	What is Circular Queue? Why Circular Queue better than Linear Queue? Write an algorithm/function for enqueue and dequeue operations on Circular Queue?	8
OR		
Q3 (a)	What is an array? How multidimensional array are stored in memory? Consider a 2-D float array A[5:10][6:10] is stored in row-wise form in the memory. Suppose the base address of an array A is 800. i) Find the total number of elements in an array A. ii) Find the address of A[7][7]	5
(b)	Write a function/algorithm to evaluate the postfix expression using stacks? Using above algorithm evaluate the following expression: 3, 4, 5, 6, *, +, 7, -, /	5
(c)	What is Deque? Write an algorithm for deletion at rear end in a Deque?	5
UNIT – II		
Q4 (a)	What is linked list? Explain advantages and disadvantages of linked list over an array? Write an algorithm/function to insert an element at the end of the linear linked list.	8
(b)	Write an algorithm/function for Quick Sort. Perform quick sort on the following values. Assume 48 as Pivot point: 48, 37, 12, 78, 19, 50, 22, 60	7
OR		
Q5 (a)	What is doubly linked list? Write a program to create a doubly linked list having information about a student such as name, roll no, marks etc. and perform insertion and deletion of a node from the beginning.	8
(b)	Write an algorithm for Radix Sort. Perform the radix sort on the following values: 380, 489, 567, 235, 323, 755, 155	7

[P.T.O.]

Q. No.	Question	Marks
UNIT – III		
Q6 (a)	What is B-tree? Compare B-tree with B+ trees. Construct B-tree of order 5 by inserting the following elements in the order of their occurrence: M, T, E, Q, K, C, G, N, H, U, W	8
(b)	The pre-order and in-order traversal of a binary tree are given below. Construct corresponding binary tree. Write its equivalent post-order traversal. Write a recursive algorithm/function for post-order traversal. Pre-order: A, B, D, C, E, G, F, H, I In-order: D, B, A, E, G, C, H, F, I	7
OR		
Q7 (a)	What is Binary Search tree? Construct a Binary Search tree for the following data in sequence 77, 33, 44, 11, 22, 55, 66 . Write its equivalent in-order traversal.	7
(b)	What kind of the Binary Search Tree is created in section (i)? Is there any associated problem with such kind of Binary Search Tree? If yes, provide the detail solution.	8
UNIT – IV		
Q8 (a)	What is minimum spanning tree? Find the minimum spanning tree for the given graph using Prim's algorithm: 	7
(b)	Explain the following collision resolution method with the help of example: a) Linear Probing b) Double hashing c) Quadratic probing d) Chaining	8
OR		
Q9 (a)	Write an algorithm for Depth First Search. Traverse the given graph using DFS. Assume vertex 1 as starting vertex: 	6
(b)	Explain briefly about the following file organization: a) Sequential access file organization b) Direct access file organization c) Indexed Sequential access file organization	9
